4(a).//QUEUE USING ARRAY PROGRAM

```
#include <stdio.h>
#define MAX 50
int queue[MAX];
int rear = - 1;
int front = - 1;

void enqueue(int data)
{
   if (rear == MAX - 1)
   printf("Queue Overflow \n");
   else
   {
      if (front == - 1)  /*If queue is initially empty */
      {
         front = 0;
                  }
      rear = rear + 1;
      queue[rear] = data;
      printf("Element inserted is : %d\n", queue[rear]);
   }
}

void dequeue()
{
   if (front == - 1 || front > rear)
   {
      printf("Queue Underflow \n");
      return ;
   }
   else
   {
      printf("\nElement deleted is : %d\n", queue[front]);
      front = front + 1;
   }
}

void display()
{
   int i;
   if (front == - 1)
      printf("Queue is empty \n");
   else
   {
      printf("Queue is : \n");
      for (i = front; i <= rear; i++)
```

```
        printf("%d ", queue[i]);
    }
} /* End of display() */
int main()
{
        enqueue(10);
        enqueue(20);
        enqueue(30);
        enqueue(50);
        enqueue(50);
        enqueue(60);

        display();

        dequeue();
        dequeue();
        display();
        return 0;
}
```

OUTPUT:

```
Element inserted is : 10
Element inserted is : 20
Element inserted is : 30
Element inserted is : 50
Element inserted is : 50
Element inserted is : 60
Queue is :
10 20 30 50 50 60
Element deleted is : 10

Element deleted is : 20
Queue is :
30 50 50 60
```